Curated Research Articles

Generated: 2026-09-07 07:00

- **Cellulose Materials Integrate Ion Transport, Interfacial Chemistry, and Structural Stability in Zinc Battery Systems** — score: 1.000 The review highlights the potential of cellulose-based materials to enhance zinc battery systems by addressing challenges such as ion transport and structural degradation. It explores how tailored cellulose structures can improve performance in both aqueous zinc-ion and alkaline zinc-air batteries through better ion conduction and interfacial contact, while also recommending priorities for future research and development. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Interfacial Engineering by a Functional Polymer Additive Ensuring Highly Reversible Zn Metal Anodes** — score: 1.000 The article presents a novel approach to enhancing the performance of aqueous zinc-ion batteries through the use of a trace polymer additive, dextran sulfate sodium. This additive forms a protective layer on the zinc anode, which modifies the solvation structure of zinc ions, improves electrode-electrolyte interface stability, and significantly boosts the reversibility of zinc plating and stripping processes. As a result, the enhanced batteries exhibit exceptional longevity and cycling performance. Journal: *Wiley: Small Methods: Table of Contents*
- **Unraveling Spatial Electrochemical Heterogeneity and Additive-Mediated Regulation in Scale-up Aqueous Zinc-Ion Pouch-Type Cells** — score: 1.000 The study explores the spatial electrochemical heterogeneity in scale-up aqueous zinc-ion pouch-type cells, revealing that different additives regulate Zn2+ transport and deposition. -cyclodextrin enhances ionic conductivity and promotes uniform ion flux, while Tween 80 stabilizes interfacial kinetics, leading to differing cycling stabilities and performance based on cell positioning. These findings provide valuable insights into optimizing zinc-ion battery design for better stability and efficiency in large-scale applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Elucidating the Suppression of High-Voltage Degradation in O3-type NaNi1/3Fe1/3Mn1/3O2: A Voltage-Switching Strategy for Na-ion Batteries** — score: 1.000 The study investigates the effects of voltage-switching on the NaNi1/3Fe1/3Mn1/3O2 cathode in sodium-ion batteries, revealing that reducing the cycling voltage from 4.3V to 4.0V effectively stabilizes previously damaged areas, halting their growth and maintaining capacity retention similar to cells only cycled at the lower voltage. This insight offers valuable understanding of managing degradation during high-depth-of-charge cycling to enhance battery longevity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Decoding Multiscale Degradation of Layered Cathodes in All-Solid-State Lithium Batteries: An Advanced Diagnostics-Driven Framework** — score: 1.000 This article review explores the intricate multiscale degradation mechanisms affecting layered oxide cathodes in all-solid-state lithium batteries, utilizing a comprehensive Diagnosis-Mechanism-Mitigation framework. It synthesizes advanced diagnostic techniques, including microscopy and spectroscopy, to elucidate both interfacial and bulk degradation processes, while also integrating machine learning tools to enhance predictive design principles for improving battery durability and performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Revealing the Dominant Role of Spatial Lithium Distribution in the Kinetics of Lithium-Ion Batteries via Operando Neutron Depth Profiling** — score: 1.000 The article explores how the spatial distribution of lithium within lithium-ion batteries significantly influences their kinetic behavior, using operando neutron depth profiling techniques. This research offers insights that could enhance the design and efficiency of energy storage materials by highlighting the importance of lithium distribution dynamics during battery operation. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Breaking the Degradation Domino in Li-Rich Layered Oxide Cathode Via Precise Electrostatic Pinning and Gradient Shielding** — score: 1.000 The article discusses advanced strategies to enhance the stability of lithium-rich layered oxide cathodes by utilizing electrostatic pinning to prevent transition metal migration and vacancy clustering, along with a gradient shielding architecture

to improve structural resilience and interface stability. These modifications result in significant improvements in capacity retention and performance during extensive cycling, addressing key challenges in lithium-ion battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Chelation Compensation Strategy for High-Performance Iron-Based Redox Flow Batteries** — score: 0.900 This article reviews the use of chelation compensation strategies to enhance the performance of iron-based redox flow batteries (IRFBs) by stabilizing metal ions through the introduction of multidentate ligands. It discusses how this approach mitigates issues of heterogeneous interface instability, side reactions, and kinetic deactivation, ultimately improving energy efficiency and cycling stability. Additionally, the review addresses the major degradation mechanisms of IRFBs and outlines current challenges and future directions for optimizing ligand design and electrolyte compatibility in large-scale energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Interfacial Stress Chemistry in Zinc Anodes for Aqueous Zinc-Ion Batteries** — score: 0.900 The article introduces a novel framework that examines the failure of zinc anodes in aqueous zinc-ion batteries through the lens of interfacial stress chemistry. It discusses how various stresses contribute to common issues like dendrite growth and mechanical degradation, and outlines strategies for regulating these stresses to improve the stability and performance of zinc anodes, thus enhancing the viability of these batteries for energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Revisiting the Closed-Pore Formation and Interfacial Micro-Environment Modulation Mechanisms of Polyacrylonitrile-Derived Carbon Nanofibers Toward Efficient Sodium Storage** — score: 0.900 This study investigates the enhancement of sodium storage in carbon nanofibers derived from polyacrylonitrile by optimizing the pyrolysis process and modulating the interfacial micro-environment. By achieving a balance between ion transport, active storage sites, and structural support through targeted microstructure evolution at 1400°C, the researchers demonstrate superior sodium storage performance with high Coulombic efficiency and stability. The findings offer valuable insights for designing advanced carbon anodes for sodium-ion technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Synergistic Gradient Interphase Engineering for Ultra-Stable and High-Rate Quasi-Solid-State Sodium Metal Batteries** — score: 0.900 This article discusses a novel gradient solid electrolyte interphase (SEI) design for quasi-solid-state sodium metal batteries (QSSMBs), which integrates a rigid inner layer with a flexible outer layer to enhance stability and ion transport. By employing a synergistic reduction mechanism with specific additives, the engineered electrolyte enables uniform sodium deposition and significantly improves the battery's performance, achieving impressive cycling stability and retention capabilities. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Multi-element surface modification of Cu2S enables oriented potassium deposition and a K2S-rich solid electrolyte interphase for ultrastable dendrite-free potassium metal batteries** — score: 0.800 The article discusses advancements in surface modification techniques for Cu2S that facilitate oriented potassium deposition and enhance the formation of a K2S-rich solid electrolyte interphase. This innovation aims to improve the stability of potassium metal batteries, specifically by preventing dendrite formation, thereby promoting longer battery life and efficiency. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Machine Learning for Rational Electrolyte Design in Lithium Batteries: Bridging Macroscopic Performance and Physically Interpretable Descriptors** — score: 0.800 This review article outlines a comprehensive framework for the rational design of lithium battery electrolytes using descriptor-driven machine learning (ML). It highlights the importance of physically interpretable descriptors in improving the predictability of ML models, addresses current challenges in the field, and proposes an integrated approach that combines virtual screening, active learning, and robotic experimentation to enhance autonomous electrolyte optimization. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Enabling Low-Temperature Fast-Charging Anode-Free Sodium Metal Batteries: A Syn-**

ergistic Design of Electrolyte Solvation and Current Collector Interface — score: 0.800 This review addresses the challenges of anode-free sodium metal batteries (AFSMBs) in achieving high energy density under low-temperature and fast-charging conditions, where sodium inventory and interfacial kinetics are significant obstacles. By integrating advancements in electrolyte solvation, current collector design, pre-sodiation methods, and artificial intelligence, a comprehensive strategy is outlined to enhance performance and stability, paving the way for practical applications that exceed 200 Wh kg−1 in extreme environments. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Nanoscale In2O3 Induced Fast Kinetics and Stable Interface for High-Loading Solid-State Cathode** — score: 0.800 The study demonstrates that incorporating nanoscale In2O3 into high-loading solid-state cathodes significantly enhances lithium-ion transport and stabilizes the interface with sulfide electrolytes, overcoming common issues associated with traditional carbon additives. This innovative approach results in a composite cathode with improved kinetic performance and enhanced cycling stability, achieving an impressive capacity retention after 500 cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Mechanistic Insights Into Entropy Regulation in Sodium/Potassium-Ion Batteries** — score: 0.800 The article reviews entropy regulation strategies in sodium-ion and potassium-ion batteries, highlighting their potential to mitigate key challenges like structural degradation and slow kinetics. It explains how medium-to-high entropy designs stabilize crystal structures and improve kinetic behaviors, thereby enhancing electrode materials and interfacial stability, while also proposing future research directions to further optimize these battery technologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Interface Engineering via Ceria Coating Reconciles Ion Transport and Lithium Compatibility in Composite Polymer Electrolytes** — score: 0.800 The study introduces a multifunctional ceria interfacial layer that enhances the performance of composite polymer electrolytes (CPEs) by improving ionic transport and preventing harmful reactions between ceramic fillers and lithium metal. This innovation results in CPEs with high ionic conductivity (0.772 mS cm−1) and demonstrates strong cycling stability in lithium-ion batteries, retaining up to 88% capacity after 750 cycles. The findings suggest a promising approach for advancing solid-state lithium-metal battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Probing far-from-equilibrium dynamics of electrical double layers** — score: 0.700 The article presents a novel approach that integrates experimental and computational techniques to investigate the behavior of ions and water at electrode interfaces. This method reveals insights into the molecular structure and dynamic evolution of electrical double layers when they are in far-from-equilibrium conditions. Journal: *Nature*
- **All-in-One 3D Printing for Highly-Stretchable Zn Metal Batteries: Resolving the Dilemma Between Mechanical Compliance and Electrochemical Stability** — score: 0.600 The article presents a novel approach to 3D printing highly stretchable zinc anodes for zinc-ion batteries by integrating material design with structural architecture. This method enhances electrochemical stability and mechanical compliance by using a flexible polyurethane-carbon nanotube network and serpentine shapes that maintain robust performance under significant tensile strain. The result is a durable and efficient battery ideal for wearable technology, demonstrating extended cycling stability and capacity retention. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Insertion-dominated anion redox in high-capacity amorphous sulfide cathodes for solid-state batteries** — score: 0.600 The article discusses a novel sulfide-based cathode active material (CAM) strategy that addresses the structural and interfacial challenges in all-solid-state batteries. By utilizing interfacial compatibility and structural disorder with anion regulation, this design allows for reversible multi-electron redox processes, resulting in enhanced specific energy, fast reaction kinetics, and prolonged cycling stability. Journal: *Joule*
- **Regulating Interfacial Electrostatics and Solvation Chemistry via a Seven-Component High-Entropy Alloy for Anode-Free Batteries** — score: 0.600 The article discusses the develop-

ment of a seven-component high-entropy alloy designed to improve the interfacial electrostatics and solvation chemistry in anode-free batteries. This innovative approach aims to enhance battery performance by optimizing the interface interactions, a key factor in increasing efficiency and lifespan. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Redox-Mediated Electrified Method for Green and Low-Cost Spent Lithium-Ion Battery Materials Recycling** — score: 0.600 The article presents a novel electrified chemical recycling method that utilizes a redox-mediated process to efficiently recover valuable materials from spent lithium-ion battery cathodes, like LiCoO2, using only electricity and water. This method not only generates useful products such as LiOH and Co(OH)2 with minimal waste but also showcases the potential for scalability to other battery materials like LiNixMnyCozO2, offering a sustainable alternative to traditional recycling techniques. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Artificial Intelligence-Assisted Investigation of Ion Transport Mechanisms and Material Screening for Solid-State Electrolytes** — score: 0.600 This article discusses the use of artificial intelligence to enhance the investigation of ion transport mechanisms and facilitate the screening process for solid-state electrolytes. The authors present advancements that could lead to improved energy storage solutions by optimizing electrolyte materials through AI-driven methodologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Rational Sulfur Vacancy Design Enhances the Electrochemical and Nonlinear Optical Properties of V2CTx/WS2** — score: 0.500 The study reports on the development of V2CTx/WS2 composites that significantly enhance both electrochemical and nonlinear optical properties through careful design of sulfur vacancies and interfacial charge transfer. The resulting material shows outstanding capacitive performance with a high specific capacity retention after extensive cycling, and impressive nonlinear optical characteristics, showcasing its potential for applications in energy storage and optical devices. Journal: *Wiley: Small Methods: Table of Contents*
- **Self-Supported Flexible Film F-Pt1/MnO2 With Long Bending Cyclability Towards Robust Wearable Al-Air Batteries** — score: 0.400 The study presents a self-supported, binder-free film F-Pt1/MnO2, specifically engineered for flexible Al-air batteries, which demonstrates remarkable bending resilience and structural integrity, maintaining performance over extensive cycling and alkaline resistance. This innovative design facilitates a peak power density of 77.9 mW cm−2, making it suitable for powering wearable electronics that require durability and consistent energy output during deformation. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Advanced Carbonate Electrolytes for Practical Lithium Metal Batteries: Challenges, Strategies and Perspectives ·** — score: 0.400 The article discusses the development of advanced carbonate electrolytes aimed at enhancing the performance of lithium metal batteries, addressing current challenges and proposing strategies for improvement. It highlights the importance of these innovations for practical applications in energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Biocatalytic closed-pore engineering in starch-derived hard carbon toward high-performance sodium-ion batteries** — score: 0.300 The article discusses advancements in the engineering of starch-derived hard carbon using biocatalytic methods, aiming to enhance the performance of sodium-ion batteries. The study demonstrates that these engineered materials can significantly improve energy storage capacities and overall battery efficiency, contributing to the development of sustainable energy solutions. Journal: *ScienceDirect Publication: Nano Energy*
- **Engineering battery architectures for multifunctionality and high performance** — score: 0.300 The article discusses the evolution of battery designs to enhance their functionality beyond energy storage, focusing on their ability to bend, stretch, and support mechanical loads. It reviews various architectural approaches, including 1D, 2D, and 3D structures, while addressing the balance between flexibility, energy density, and manufacturing challenges. The authors emphasize the need for advanced computational design and scalable production techniques to transition from prototypes to practical applications. Journal: *Joule*

- **Surface-dominant pseudocapacitance in graphene@hard carbon heterostructure for fast and durable sodium storage** — score: 0.300 The article presents a novel graphene@hard carbon heterostructure that exhibits surface-dominant pseudocapacitance, enhancing the performance of sodium storage systems. This design promotes rapid charge and discharge cycles while ensuring durability, making it a promising candidate for efficient energy storage applications. Journal: *ScienceDirect Publication: Nano Energy*
- **Synergistic Cation-Anion Chemistry to Modulate Active Water Molecules for Highly Stable Zinc Metal Anodes** — score: 0.200 The article discusses the innovative approach of using synergistic cation-anion chemistry to enhance the stability of zinc metal anodes by effectively modulating active water molecules. This advancement addresses challenges related to zinc anode performance, promising improved longevity and efficiency in energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Synergistic Dual-Ion Engineering Strategy Enhances Ion Transport and High-Voltage Stability for Halide Solid-State Electrolytes** — score: 0.200 The article presents a novel dual-ion engineering strategy aimed at improving ion transport and enhancing the high-voltage stability of halide solid-state electrolytes. This approach could significantly advance the performance and safety of energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **High-Voltage, Ultra-Long Cycling Alkaline AIFBs Enabled by a Rationally Designed Six-Coordinate Fe Complex** — score: 0.200 The article presents a novel alkaline anion exchange membrane fuel cell (AIFB) that utilizes a specifically designed six-coordinate iron complex, achieving high voltage and remarkably extended cycling stability. This advancement highlights the potential for enhanced performance in energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Kilogram-Scale Spherical MnO2 via Structure–Surface Dual Engineering for Wide-Temperature Aqueous Zinc-Ion Batteries** — score: 0.200 The article discusses the development of a kilogram-scale spherical manganese dioxide (MnO2) through a dual engineering approach, enhancing both its structure and surface properties. This advancement is aimed at improving the performance of aqueous zinc-ion batteries across a wide temperature range, potentially offering innovative solutions for energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Supramolecular Engineering of Nanocellulose/COF Composite Membranes: Reconciling Multidimensional Proton Pathways With Framework Anchoring** — score: 0.200 The article presents a novel membrane created by integrating sulfonated nanocellulose with a 2D covalent organic framework, enhancing proton conductivity while minimizing swelling. This supramolecular design results in impressive proton conductivities and improved fuel-cell performance, significantly outperforming traditional membranes in both stability and efficiency under varying humidity conditions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Strain Engineering in High-Entropy Electrocatalysts: Mechanisms, Strategies and Water Splitting Applications** — score: 0.200 This review article focuses on the innovative approach of strain engineering in high-entropy electrocatalysts to enhance their performance in water splitting applications. It discusses the mechanisms and strategies for modulating these materials' atomic structures to optimize their electronic characteristics and reaction kinetics, while also highlighting the importance of advanced characterization techniques and AI in developing efficient catalysts for electrochemical reactions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Accelerated Oxygen Evolution Kinetics in Layered Double Perovskites via Hydroxide Surface Engineering for Anion Exchange Membrane Water Electrolysis (Adv. Energy Mater. 33/2026)** — score: 0.200 The study reveals that a NiFe(OH)x coating enhances the performance of layered double perovskites for oxygen evolution in anion exchange membrane water electrolysis. This modification facilitates improved electron transfer and stabilizes cation dynamics, resulting in high catalytic activity and remarkable durability over 1000 hours. Journal: *Wiley: Advanced Energy Materials:*

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- **Chemical Ex-Solution of Bi3-xTaO7 Support for Electrical Integration at the Interface With Ir Bi Alloy Electrocatalysts in Acidic Oxygen Evolution Reaction** — score: 0.200 The article presents a novel approach to enhance the performance of electrocatalysts in water electrolysis by partially exsolving bismuth (Bi) from a Bi3TaO7 support. This process results in the formation of active Ir Bi alloy domains and a conductive Bi-deficient support, significantly improving interfacial electronic properties and thereby enhancing both half-cell and single-cell oxygen evolution reaction activities while reducing iridium (Ir) loading. The findings suggest a promising strategy for developing efficient, durable electrocatalysts suitable for proton exchange membrane water electrolysis. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Clean Tin Oxide via Volatile Additive Engineering for High-performance Perovskite Photovoltaics** — score: 0.000 The study introduces a volatile additive engineering method using pyruvic acid to enhance the quality of SnO2 surfaces in perovskite solar cells. By reducing residual groups and oxygen-related defects through effective removal during annealing, this approach significantly improves charge recombination and thermal stability, resulting in lab-scale efficiencies of 26.17% and maintaining over 80% efficiency after extended operation. The method is scalable, achieving competitive performance in larger solar modules as well. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Weakening Local Chemical Bonds for Enhanced Thermoelectric Performance in YbMg2Sb2 Zintl Compounds** — score: 0.000 The article explores how alloying YbMg2Sb2 with isovalent Zn or Cd significantly lowers lattice thermal conductivity beyond traditional models by weakening local chemical bonds, which negatively affects phonon propagation. This modification leads to a remarkable thermoelectric figure of merit (ZT) of 1.4 at 873 K for the Cd-alloyed variant, emphasizing the importance of chemical bond manipulation in developing materials with superior thermoelectric performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Proton Flux Engineering for Selective CO2 Electroreduction** — score: 0.000 This article presents a comprehensive framework for proton flux engineering in the electroreduction of CO2, emphasizing the critical role of protons in facilitating selective and efficient product formation. By linking proton dynamics with electron transfer processes and interfacial interactions, the authors propose a multiscale approach that integrates various factors such as transport pathways and local electric fields to enhance the conversion of CO2 into desired carbon products while minimizing hydrogen evolution. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Overcoming the interfacial heterogeneity of pyramidal structure for efficient perovskite/silicon tandem solar cells** — score: 0.000 The article discusses the use of an asymmetric phosphonic acid-based self-assembled monolayer (L30) to enhance the performance of perovskite/silicon tandem solar cells by creating a uniform coating over submicron-textured silicon substrates. This innovative approach reduces interfacial heterogeneity, leading to a significant increase in efficiency, with a recorded stabilized efficiency of 34.15% in the top-performing solar cell. Journal: *Joule*
- **Concentration Polarization-Regulated Interfacial Evolution in High-Rate Discharging Anode-Free Lithium Metal Batteries** — score: 0.000 The article discusses the phenomenon of concentration polarization and its impact on the interfacial dynamics in anode-free lithium metal batteries during high-rate discharge. The authors investigate how this polarization affects lithium deposition and overall battery performance, aiming to enhance the efficiency and longevity of these energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **A flexible ionic gel electrode-enabled TENG–EMG proximity–tactile sensor for vision-free robotic grasping and object recognition** — score: 0.000 This article discusses the development of a versatile sensor combining a triboelectric nanogenerator (TENG) and an electrochemical mechanical gauge (EMG) using a flexible ionic gel electrode. This innovative sensor enables vision-free robotic systems to effectively grasp and recognize objects by detecting proximity and tactile feedback, paving the way for advanced robotic applications. Journal: *ScienceDirect Publication: Nano Energy*

- **Sustainable cellulose nanofibrils as ultrasound-driven piezoelectric interfaces for wireless neuromodulation** — score: 0.000 The article discusses the development of sustainable cellulose nanofibrils that function as piezoelectric interfaces, utilizing ultrasound to enable wireless neuromodulation. This innovative approach could advance neurotechnology by facilitating non-invasive modulation of neural activity. Journal: *ScienceDirect Publication: Nano Energy*
- **Improvement in the Thermoelectric Performance of Manganese Telluride With the Addition of Cu2Se** — score: 0.000 The addition of 8 at.% Cu2Se to manganese telluride (MnTe) significantly enhances its thermoelectric performance by increasing carrier concentration and reducing thermal conductivity through phonon scattering, achieving a maximum ZT value of approximately 0.72 at 873 K. This improvement is attributed to the unique behavior of Cu-ions in Cu2Se at elevated temperatures, which facilitates better electrical properties while lowering thermal transport. Journal: *Wiley: Small Methods: Table of Contents*
- **Transfer-Stacked and Pre-Tunable Bilayer Photoresist for High-Yield Dry Lift-off and Multiscale Patterning** — score: 0.000 The paper introduces an innovative bilayer photoresist technique that utilizes a single material to achieve solvent-free, high-yield dry lift-off in photolithography. By employing surface modification and UV pre-exposure, it offers precise control over undercut structures and compatibility with processes like sputtering, enabling robust multiscale patterning from wafer-scale down to 17 nm features. This method demonstrates significant potential for applications in optoelectronics, including displays and micro-LEDs. Journal: *Wiley: Small Methods: Table of Contents*
- **Correction to "Cell-Free Protein Synthesis in Porous Parylene Scaffolds"** — score: 0.000 This article presents a correction to previously published work on cell-free protein synthesis conducted within porous parylene scaffolds. It addresses key methodological details and results, ensuring accurate interpretation of data regarding the efficiency and application of the scaffolds in protein production. Journal: *Wiley: Small Methods: Table of Contents*
- **A Unified Review of Altermagnetism: Momentum-Space Spin Polarization and Emerging Applications** — score: 0.000 This review provides a comprehensive examination of altermagnetism, a novel class of collinear magnetism characterized by zero net magnetization but momentum-dependent spin polarization, stemming from intricate crystal symmetries. It covers the theoretical basis, experimental identification methods, and various material types, while exploring implications for future technologies in spintronics, superconductors, and quantum devices. Journal: *Wiley: Small Methods: Table of Contents*
- **A Plasmonic Nanopore Sensor for Dual-Gating Optoelectronic Detection of Cysteine** — score: 0.000 The article presents a dual-mode optoelectronic plasmonic nanopore sensor specifically designed for the sensitive and selective detection of cysteine, leveraging a custom fluorescent probe. By combining plasmon-enhanced fluorescence with electrochemical signals, the sensor achieves enhanced accuracy and minimal false responses, successfully quantifying cysteine in living cancer cells and demonstrating adaptability for detecting various small molecules. Journal: *Wiley: Small Methods: Table of Contents*
- **Donor–Acceptor Charge-Selective Materials for High-Performance Perovskite Solar Cells: Molecular Design, Structure–Property Correlation and Device Performance** — score: 0.000 This review focuses on the advancement of donor-acceptor charge-selective materials in perovskite solar cells, which are vital for enhancing device efficiency and stability. It examines how the molecular structure and properties of these materials influence interfacial dynamics and overall device performance, addressing the challenges faced by conventional organic charge-selective materials. Future research directions for optimizing these materials towards commercial viability are also outlined. Journal: *Wiley: Advanced Energy Materials: Table of Contents*